



Southeast University

Department of Computer Science & Engineering

Computer Networking Project

CEP

Submitted by,

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Section – 19

Course Code – CSE342

Course Title – Computer Networking Lab

Submitted to,

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1. Objectives

Base IPv4 block: 172.25.0.0/23 | Router-to-router link size: 4 addresses (/30) | VLANs: VLAN 10 (30 hosts), VLAN 30 (25 hosts), VLAN 50 (10 hosts) | NAT router LAN: 20 hosts.

1. Design a Variable Length Subnet Mask (VLSM) addressing plan for the six-router enterprise network (Router0 to Router5), the two-switch VLAN segment and the NAT-PAT border, using only the assigned 172.25.0.0/23 block.
2. Configure DHCP so that every PC in every router LAN automatically receives an IP address, subnet mask, default gateway and DNS server address.
3. Deploy two DNS servers, each alongside three web servers hosting distinct websites, and verify that domain names resolve correctly from client PCs.
4. Configure the OSPF routing protocol on all six core routers so every LAN can reach every other LAN without static routes.
5. Build a two-switch VLAN architecture (Switch6 and Switch7) with three VLANs, using access and trunk ports, so that devices communicate only inside their own VLAN.
6. Implement NAT-PAT on the two-router border setup (Router6 and Router7) so that many private addresses share one address when leaving the private network.

2. Introduction

This project models the network of a multi-branch enterprise with six core routers (Router0 to Router5). Each router serves a local area network with its own switch (Switch0 to Switch5), a DHCP server and three end-user PCs. The LANs are sized for 65, 25, 15, 10, 10 and 5 hosts, which is why one fixed subnet size would waste addresses; VLSM lets each network take a block that fits its actual need.

Four services are layered on top of the addressing plan. DHCP removes manual host configuration. DNS with web servers gives name-based access to internal websites. OSPF, a link-state routing protocol, lets the six routers learn each other's networks automatically. A separate two-switch segment demonstrates VLAN isolation at Layer 2, and a pair of border routers demonstrates NAT-PAT.

The design follows the model Packet Tracer file supplied for the course, and the device names in this report are the same as the names in that file. The file has three canvases: the Base Setup (Router0 to Router5 connected in a chain), the VLAN Setup (Switch6 and Switch7) and the NAT-PAT Setup (Router6 and Router7). They are shown below as Figure 1, Figure 2 and Figure 3. Every address comes from the group's assigned block, 172.25.0.0/23.

3. Implementation

3.1 Network Topology and Assumptions

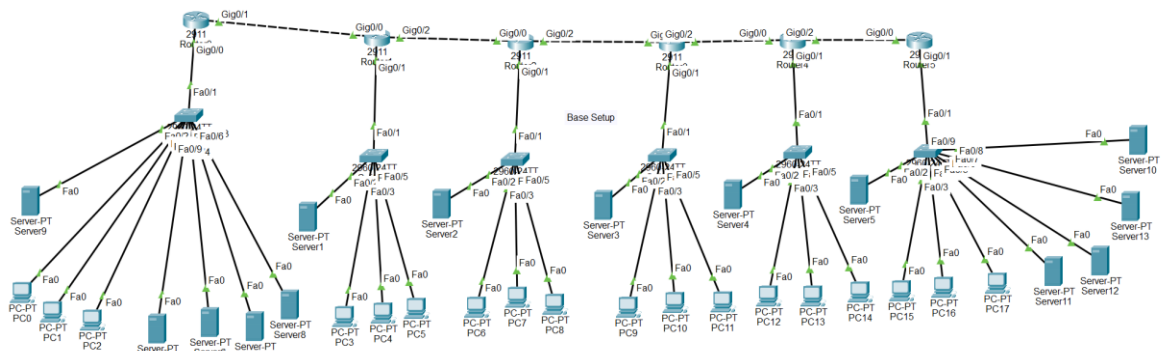


Figure 1 - Base Setup: Router0 to Router5 with their switches, DHCP servers, DNS servers, web servers and PCs

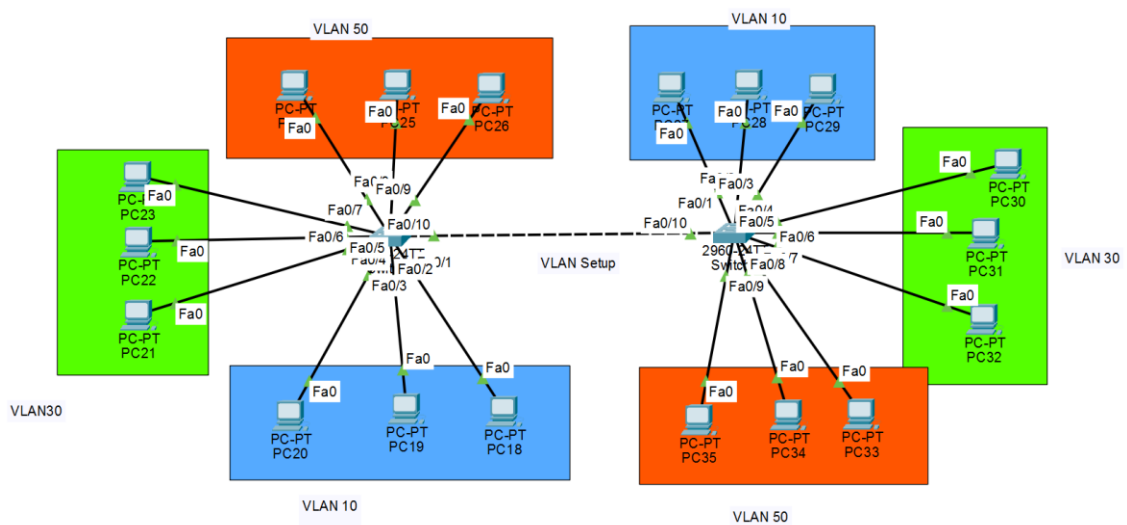


Figure 2 - VLAN Setup: Switch6 and Switch7 trunked together, carrying three VLANs (blue, green, orange)

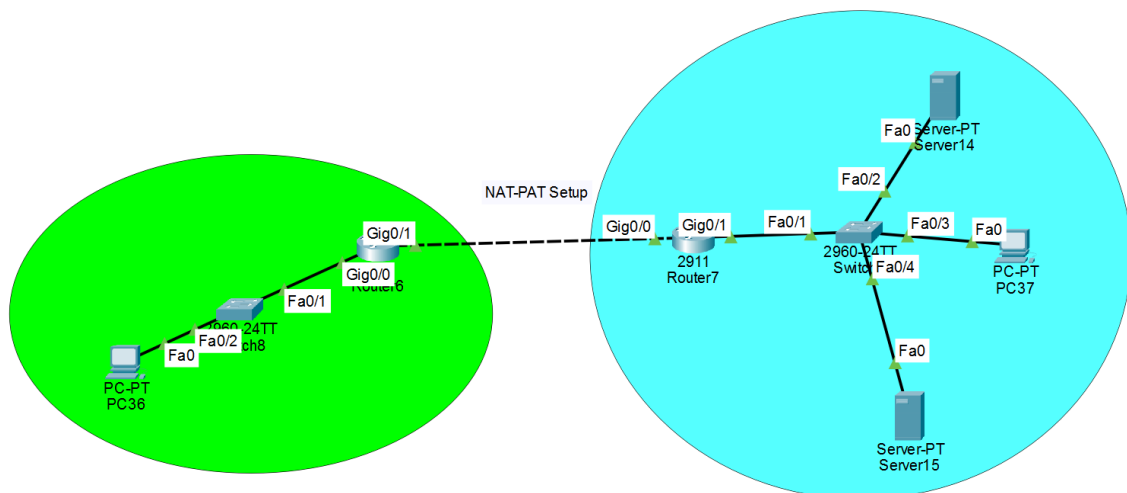


Figure 3 - NAT-PAT Setup: Router6 (inside, private) connected to Router7 (outside)

Devices in the Packet Tracer file

Section	Devices	Notes
Base Setup	Router0 to Router5, Switch0 to Switch5	Routers joined in a chain: 0-1, 1-2, 2-3, 3-4, 4-5 (five links)

Section	Devices	Notes
Router0 LAN	Server0, Server6, Server7, Server8, Server9, PC0-PC2	Five servers: DHCP, DNS and three web servers
Router1 LAN	Server1, PC3-PC5	DHCP server + 3 PCs
Router2 LAN	Server2, PC6-PC8	DHCP server + 3 PCs
Router3 LAN	Server3, PC9-PC11	DHCP server + 3 PCs
Router4 LAN	Server4, PC12-PC14	DHCP server + 3 PCs
Router5 LAN	Server5, Server10, Server11, Server12, Server13, PC15-PC17	Five servers: DHCP, DNS and three web servers
VLAN Setup	Switch6, Switch7, PC18-PC35	Three PCs of each colour on each switch (18 PCs)
NAT-PAT Setup	Router6, Switch8, PC36 (inside); Router7, Switch9, Server14, Server15, PC37 (outside)	Router6 is the private side, Router7 the outside side

Assumptions

- The project brief refers to routers "R1 to R6". In this report R1 to R6 are the Packet Tracer routers Router0 to Router5 in that order, so R1 = Router0, R2 = Router1, and so on up to R6 = Router5. The host sizes for G8 (65, 25, 15, 10, 10, 5) are applied in this order.
- The six core routers are cabled in a chain, exactly as in the Packet Tracer file, which gives five router-to-router links. The brief asks for six link blocks, so a sixth is reserved for a Router5-Router0 link that closes the chain into a ring. It is documented in the addressing table but not cabled or advertised, because the supplied topology does not contain it.
- Each router LAN has one Server-PT acting as its DHCP server, so DHCP is configured on the servers. Router6 is the exception: its LAN has no server, so the router provides DHCP itself.
- DNS and web servers sit on the two end LANs. Router0 LAN: Server0 = DHCP, Server6 = DNS Server1, Server7, Server8 and Server9 = Web Server1, 2 and 3. Router5 LAN: Server5 = DHCP, Server10 = DNS Server2, Server11, Server12 and Server13 = Web Server4, 5 and 6. These two LANs are sized larger to hold their servers.
- The VLAN segment is self-contained: two switches and no router, which is what enforces the rule that devices communicate only inside their own VLAN. The three colours in the canvas are used as VLAN 10 = blue, VLAN 30 = green and VLAN 50 = orange.
- For NAT-PAT, Router6's inside LAN (Switch8, PC36) is the private network. Router7 and its LAN (Switch9, Server14, Server15, PC37) represent the outside network. Because the brief requires every block to be derived from the assigned range, the outside side also uses addresses from 172.25.0.0/23 and is treated as the "public" side of the lab.

3.2 VLSM and IP Addressing (Question 1)

Requirements and block sizes

For each network, the number of addresses needed is the host count plus the extra devices that live in the same subnet (router gateway, DHCP server, and DNS/web servers where present), plus the network and broadcast addresses. The smallest power-of-two block that fits is chosen.

Network	Hosts given	Extra devices in subnet	Total needed	Block
Router0 LAN (R1)	65	Gateway, Server0 DHCP, Server6 DNS, Server7-9 web (6)	$65+6+2 = 73$	/25 (128)
Router1 LAN (R2)	25	Gateway, Server1 DHCP (2)	$25+2+2 = 29$	/27 (32)
Router2 LAN (R3)	15	Gateway, Server2 DHCP (2)	$15+2+2 = 19$	/27 (32)

Network	Hosts given	Extra devices in subnet	Total needed	Block
Router3 LAN (R4)	10	Gateway, Server3 DHCP (2)	10+2+2 = 14	/28 (16)
Router4 LAN (R5)	10	Gateway, Server4 DHCP (2)	10+2+2 = 14	/28 (16)
Router5 LAN (R6)	5	Gateway, Server5 DHCP, Server10 DNS, Server11-13 web (6)	5+6+2 = 13	/28 (16)
VLAN 10	30	None (no router)	30+2 = 32	/27 (32)
VLAN 30	25	None (no router)	25+2 = 27	/27 (32)
VLAN 50	10	None (no router)	10+2 = 12	/28 (16)
NAT inside LAN (Router6)	20	Gateway (1)	20+1+2 = 23	/27 (32)
NAT outside LAN (Router7)	not given	Gateway, Server14, Server15, PC37 (4)	4+2 = 6	/29 (8)
6 router links	2 each	None	4 each	/30 (4) x 6
NAT link (Router6-Router7)	2	None	4	/30 (4)

Blocks were allocated largest first (one /25, five /27, four /28, one /29, seven /30). Allocating in descending size guarantees that every block begins on a valid boundary for its size and leaves no gaps between blocks.

IP addressing table

Network	Network address	Subnet mask	Broadcast	Usable range	Default gateway
Router0 LAN	172.25.0.0	255.255.255.128 (/25)	172.25.0.127	172.25.0.1 - 172.25.0.126	172.25.0.1
Router1 LAN	172.25.0.128	255.255.255.224 (/27)	172.25.0.159	172.25.0.129 - 172.25.0.158	172.25.0.129
Router2 LAN	172.25.0.160	255.255.255.224 (/27)	172.25.0.191	172.25.0.161 - 172.25.0.190	172.25.0.161
VLAN 10	172.25.0.192	255.255.255.224 (/27)	172.25.0.223	172.25.0.193 - 172.25.0.222	None (L2 only)
VLAN 30	172.25.0.224	255.255.255.224 (/27)	172.25.0.255	172.25.0.225 - 172.25.0.254	None (L2 only)
NAT inside LAN (Router6)	172.25.1.0	255.255.255.224 (/27)	172.25.1.31	172.25.1.1 - 172.25.1.30	172.25.1.1
Router3 LAN	172.25.1.32	255.255.255.240 (/28)	172.25.1.47	172.25.1.33 - 172.25.1.46	172.25.1.33
Router4 LAN	172.25.1.48	255.255.255.240 (/28)	172.25.1.63	172.25.1.49 - 172.25.1.62	172.25.1.49
Router5 LAN	172.25.1.64	255.255.255.240 (/28)	172.25.1.79	172.25.1.65 - 172.25.1.78	172.25.1.65
VLAN 50	172.25.1.80	255.255.255.240 (/28)	172.25.1.95	172.25.1.81 - 172.25.1.94	None (L2 only)
NAT outside LAN (Router7)	172.25.1.96	255.255.255.248 (/29)	172.25.1.103	172.25.1.97 - 172.25.1.102	172.25.1.97
Link Router0-Router1	172.25.1.104	255.255.255.252 (/30)	172.25.1.107	172.25.1.105 - 172.25.1.106	-
Link Router1-Router2	172.25.1.108	255.255.255.252 (/30)	172.25.1.111	172.25.1.109 - 172.25.1.110	-
Link Router2-Router3	172.25.1.112	255.255.255.252 (/30)	172.25.1.115	172.25.1.113 - 172.25.1.114	-
Link Router3-Router4	172.25.1.116	255.255.255.252 (/30)	172.25.1.119	172.25.1.117 - 172.25.1.118	-

Network	Network address	Subnet mask	Broadcast	Usable range	Default gateway
Link Router4-Router5	172.25.1.120	255.255.255.252 (/30)	172.25.1.123	172.25.1.121 - 172.25.1.122	-
Link Router5-Router0 (reserved, not cabled)	172.25.1.124	255.255.255.252 (/30)	172.25.1.127	172.25.1.125 - 172.25.1.126	-
NAT link Router6-Router7	172.25.1.128	255.255.255.252 (/30)	172.25.1.131	172.25.1.129 - 172.25.1.130	-

Address usage: the plan spans 172.25.0.0 to 172.25.1.131, which is $128 + (5 \times 32) + (4 \times 16) + 8 + (7 \times 4) = 388$ of the 512 addresses in the /23 block (about 76%). The remaining range, 172.25.1.132 to 172.25.1.255 (124 addresses), is free for future growth. Every network in the project, including the NAT-PAT segment, is taken from the assigned block.

Router interface addressing

Router	Gi0/0	Gi0/1	Gi0/2
Router0	172.25.0.1/25 (LAN)	172.25.1.105/30 (to Router1)	not used
Router1	172.25.1.106/30 (to Router0)	172.25.0.129/27 (LAN)	172.25.1.109/30 (to Router2)
Router2	172.25.1.110/30 (to Router1)	172.25.0.161/27 (LAN)	172.25.1.113/30 (to Router3)
Router3	172.25.1.114/30 (to Router2)	172.25.1.33/28 (LAN)	172.25.1.117/30 (to Router4)
Router4	172.25.1.118/30 (to Router3)	172.25.1.49/28 (LAN)	172.25.1.121/30 (to Router5)
Router5	172.25.1.122/30 (to Router4)	172.25.1.65/28 (LAN)	not used (172.25.1.125/30 reserved for ring)
Router6	172.25.1.1/27 (inside LAN)	172.25.1.129/30 (to Router7)	not used
Router7	172.25.1.130/30 (to Router6)	172.25.1.97/29 (outside LAN)	not used

Router-to-router cabling (from Figure 1)

Link	One end	Other end	Subnet
Router0 - Router1	Router0 Gi0/1	Router1 Gi0/0	172.25.1.104/30
Router1 - Router2	Router1 Gi0/2	Router2 Gi0/0	172.25.1.108/30
Router2 - Router3	Router2 Gi0/2	Router3 Gi0/0	172.25.1.112/30
Router3 - Router4	Router3 Gi0/2	Router4 Gi0/0	172.25.1.116/30
Router4 - Router5	Router4 Gi0/2	Router5 Gi0/0	172.25.1.120/30
Each router - its switch	Router0 Gi0/0; Router1-5 Gi0/1	Switch0-5 Fa0/1	Router LAN

Static server addresses (excluded from DHCP pools)

LAN	Device	IP address	DHCP pool for PCs
Router0	Gateway / Server0 (DHCP) / Server6 (DNS Server1)	172.25.0.1 / .2 / .3	172.25.0.7 onward, 65 users (PC0-PC2)
Router0	Server7 / Server8 / Server9 (Web Server1 / 2 / 3)	172.25.0.4 / .5 / .6	
Router1	Gateway / Server1 (DHCP)	172.25.0.129 / .130	172.25.0.131 onward, 25 users (PC3-PC5)
Router2	Gateway / Server2 (DHCP)	172.25.0.161 / .162	172.25.0.163 onward, 15 users (PC6-PC8)
Router3	Gateway / Server3 (DHCP)	172.25.1.33 / .34	172.25.1.35 onward, 10 users (PC9-PC11)
Router4	Gateway / Server4 (DHCP)	172.25.1.49 / .50	172.25.1.51 onward, 10 users (PC12-PC14)

LAN	Device	IP address	DHCP pool for PCs
Router5	Gateway / Server5 (DHCP) / Server10 (DNS Server2)	172.25.1.65 / .66 / .67	172.25.1.71 onward, 5 users (PC15-PC17)
Router5	Server11 / Server12 / Server13 (Web Server4 / 5 / 6)	172.25.1.68 / .69 / .70	

3.3 DHCP Configuration (Question 2)

Each LAN in Figure 1 has a dedicated DHCP server. On every server the IP configuration was set statically, then the DHCP service was switched on with the values below. The first addresses of each subnet are reserved for the gateway and the static servers, so the pool starts after them.

LAN	DHCP server	Gateway issued	DNS issued	Pool start	Mask	Max users
Router0	Server0 172.25.0.2	172.25.0.1	172.25.0.3	172.25.0.7	255.255.255.128	65
Router1	Server1 172.25.0.130	172.25.0.129	172.25.0.3	172.25.0.131	255.255.255.224	25
Router2	Server2 172.25.0.162	172.25.0.161	172.25.0.3	172.25.0.163	255.255.255.224	15
Router3	Server3 172.25.1.34	172.25.1.33	172.25.1.67	172.25.1.35	255.255.255.240	10
Router4	Server4 172.25.1.50	172.25.1.49	172.25.1.67	172.25.1.51	255.255.255.240	10
Router5	Server5 172.25.1.66	172.25.1.65	172.25.1.67	172.25.1.71	255.255.255.240	5

Router0 to Router2 hand out DNS Server1 (172.25.0.3) and Router3 to Router5 hand out DNS Server2 (172.25.1.67), so each half of the network uses the nearer DNS server. Clients broadcast their DHCP requests inside their own LAN, so no relay agent is needed because every LAN has its own server. The three PCs in each LAN (PC0-PC2, PC3-PC5, and so on) were set to DHCP and receive the first addresses of the pool.

The NAT inside LAN (Router6, Switch8, PC36) has no separate server, so Router6 provides DHCP itself:

```
Router6(config)# ip dhcp excluded-address 172.25.1.1
Router6(config)# ip dhcp pool NATLAN
Router6(dhcp-config)# network 172.25.1.0 255.255.255.224
Router6(dhcp-config)# default-router 172.25.1.1
Router6(dhcp-config)# dns-server 172.25.1.98
```

Verification: each PC was set to DHCP under Desktop > IP Configuration, and the received address, mask, gateway and DNS server were checked with ipconfig /all.

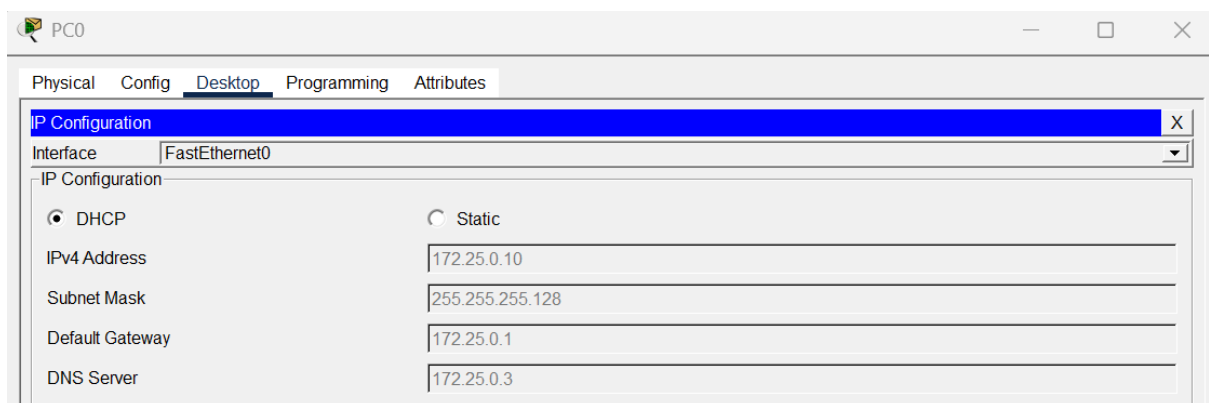


Figure 4 - PC in the Router0 LAN receiving address, mask, gateway and DNS by DHCP

3.4 DNS and Web Server Deployment (Question 3)

Two DNS servers and six web servers were deployed, with three web servers next to each DNS server. Each web server hosts a distinct site whose index.html was edited to show a different page. On each DNS server the DNS service was enabled and one A record was added per website.

Server	IP address	Website / role	Location
Server6 (DNS Server1)	172.25.0.3	Resolves the sites below	Router0 LAN
Server7 (Web Server1)	172.25.0.4	seu.edu.bd	Router0 LAN
Server8 (Web Server2)	172.25.0.5	ums.seu.edu.bd	Router0 LAN
Server9 (Web Server3)	172.25.0.6	www.google.com	Router0 LAN
Server10 (DNS Server2)	172.25.1.67	Resolves the sites below	Router5 LAN
Server11 (Web Server4)	172.25.1.68	hexlen.com	Router5 LAN
Server12 (Web Server5)	172.25.1.69	facebook.com	Router5 LAN
Server13 (Web Server6)	172.25.1.70	x.com	Router5 LAN

Each DNS server holds A records for all six websites (its own three plus the other cluster's three), so a client can open any site no matter which DNS server it received from DHCP. Only the first three records on each server point to web servers in the same LAN; the other three point across the OSPF backbone.

Verification: from a PC, the web browser was used to open each site by domain name (not by IP), and nslookup was run to show the resolved address.

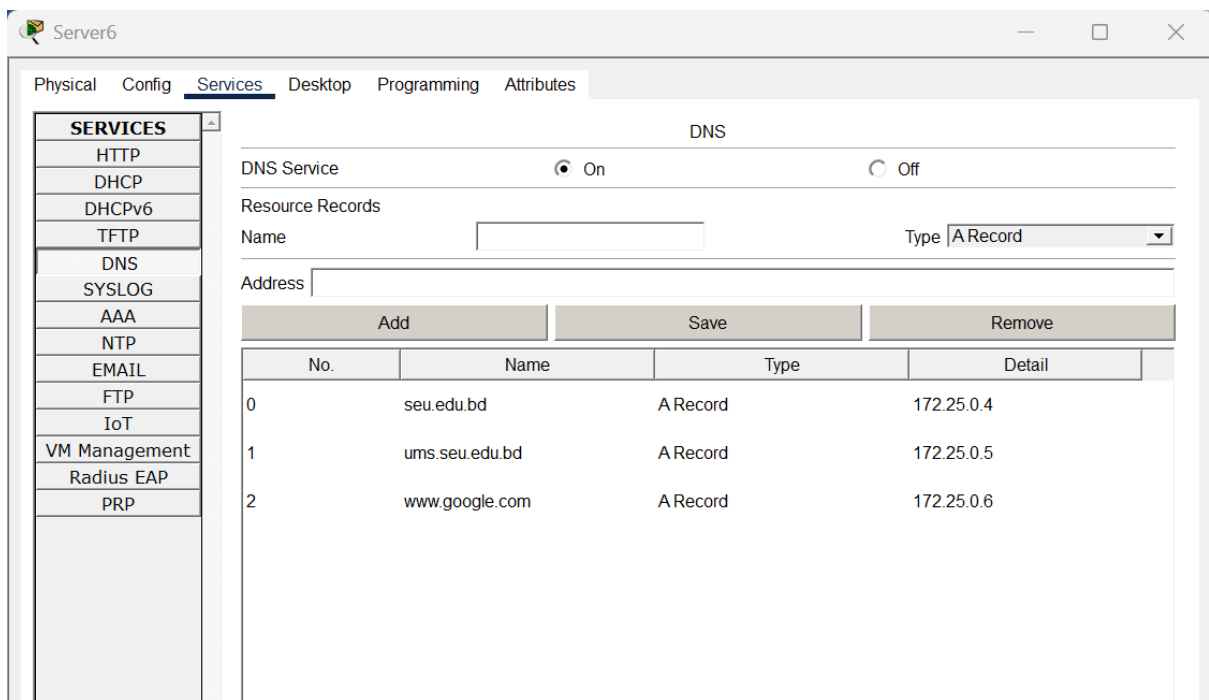


Figure 5 - DNS Server1 (Server6) A records

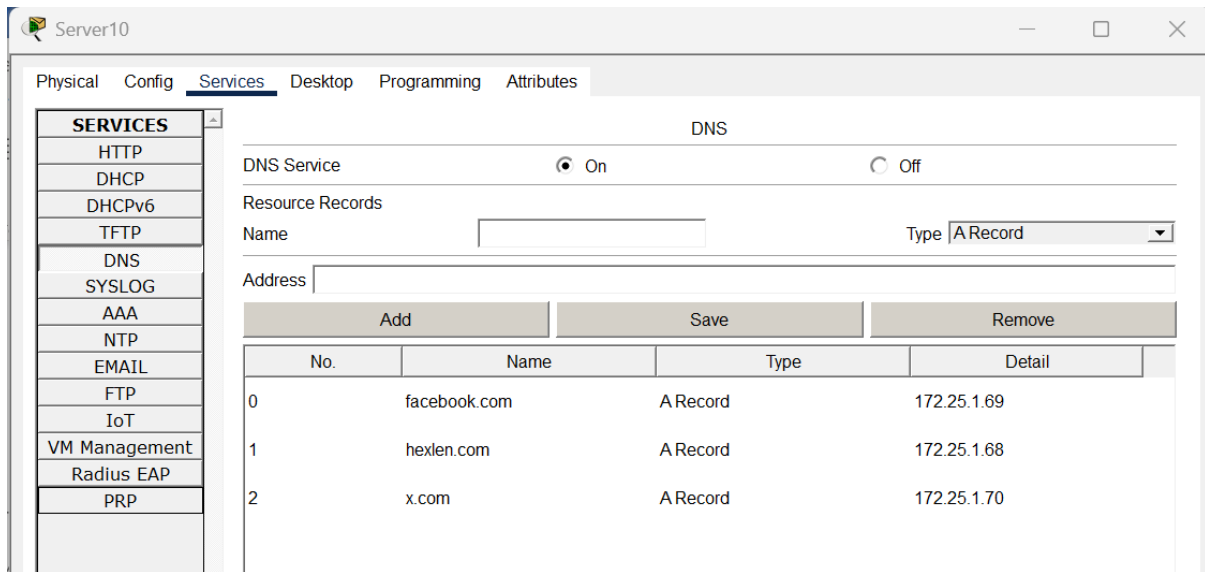


Figure 6 - DNS Server2 (Server10) A records

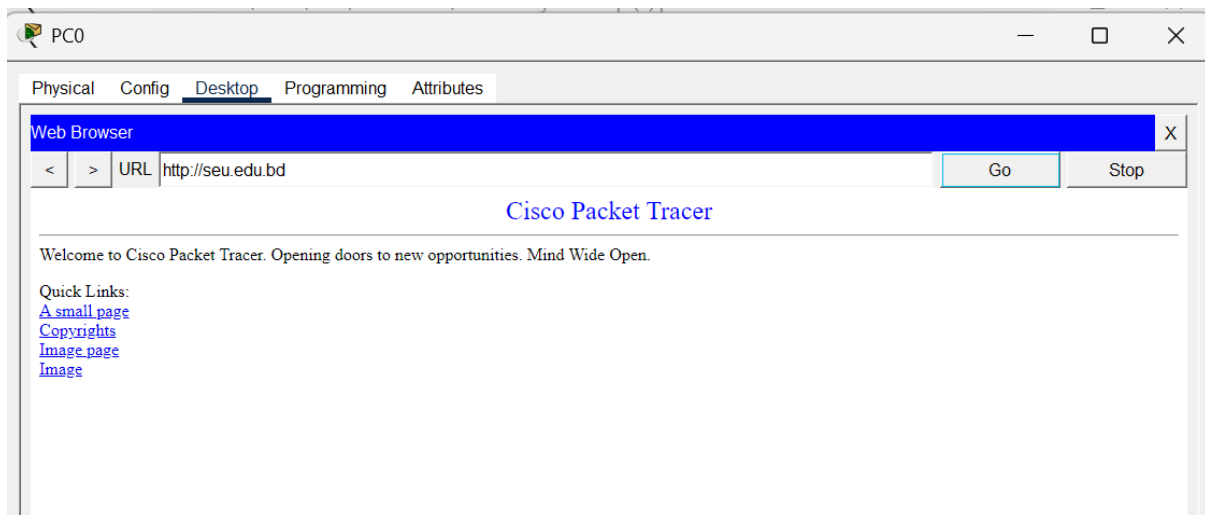


Figure 7 - Browser on a PC loading a website by domain name

3.5 OSPF Dynamic Routing (Question 4)

OSPF process 1 runs in area 0 on all six core routers. Each router advertises its LAN and the backbone links it is cabled to. Because the routers are in a chain, Router0 and Router5 have one OSPF neighbour and Router1 to Router4 have two. A network statement uses a wildcard mask, which is the inverse of the subnet mask (for example 255.255.255.128 becomes 0.0.0.127).

Router0 (complete example)

```
Router0(config)# router ospf 1
Router0(config-router)# network 172.25.0.0 0.0.0.127 area 0
Router0(config-router)# network 172.25.1.104 0.0.0.3 area 0
```

Router	network statements (all area 0)
Router0	172.25.0.0 0.0.0.127 172.25.1.104 0.0.0.3
Router1	172.25.0.128 0.0.0.31 172.25.1.104 0.0.0.3 172.25.1.108 0.0.0.3
Router2	172.25.0.160 0.0.0.31 172.25.1.108 0.0.0.3 172.25.1.112 0.0.0.3
Router3	172.25.1.32 0.0.0.15 172.25.1.112 0.0.0.3 172.25.1.116 0.0.0.3
Router4	172.25.1.48 0.0.0.15 172.25.1.116 0.0.0.3 172.25.1.120 0.0.0.3

Router	network statements (all area 0)
Router5	172.25.1.64 0.0.0.15 172.25.1.120 0.0.0.3

Router6 and Router7 are not part of the OSPF domain. They form a separate border setup that uses a default route instead. If the reserved Router5-Router0 link were cabled, 172.25.1.124 0.0.0.3 would be added on both routers and the chain would become a ring with a redundant path.

Verification: show ip ospf neighbor confirms one full neighbour on Router0 and Router5 and two on Router1 to Router4, show ip route ospf lists the learned networks, and pings between PCs in distant LANs (for example PC0 to PC17) confirm end-to-end reachability.

```
C:\>ping 172.25.0.165

Pinging 172.25.0.165 with 32 bytes of data:

Request timed out.
Reply from 172.25.0.165: bytes=32 time=1ms TTL=125
Reply from 172.25.0.165: bytes=32 time<1ms TTL=125
Reply from 172.25.0.165: bytes=32 time<1ms TTL=125

Ping statistics for 172.25.0.165:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Figure 8 - Successful pings between PCs in different router LANs

3.6 VLAN Architecture and Traffic Isolation (Question 5)

Two switches (Switch6 and Switch7) carry three VLANs. On each switch, three access ports belong to each VLAN, and a trunk port connects the two switches so each VLAN spans both. There is no router in this segment, so nothing can forward packets between VLANs.

VLAN	Colour / name	Subnet	PCs on Switch6 / Switch7	PC addresses
10	Blue / VLAN10	172.25.0.192/27	PC18-20 / PC27-29	172.25.0.193 - .198 (PC18-20 = .193-.195, PC27-29 = .196-.198)
30	Green / VLAN30	172.25.0.224/27	PC21-23 / PC30-32	172.25.0.225 - .230 (PC21-23 = .225-.227, PC30-32 = .228-.230)
50	Orange / VLAN50	172.25.1.80/28	PC24-26 / PC33-35	172.25.1.81 - .86 (PC24-26 = .81-.83, PC33-35 = .84-.86)
Trunk	-	Carries 10, 30, 50	Link between Switch6 and Switch7	-

The PCs are addressed manually (there is no DHCP server here) with the subnet mask of their VLAN and no default gateway. Access ports Fa0/1-3 are placed in VLAN 10, Fa0/4-6 in VLAN 30 and Fa0/7-9 in VLAN 50, with Fa0/10 as the trunk. Both switches use the same configuration:

```
Switch(config)# vlan 10
Switch(config-vlan)# name VLAN10
Switch(config)# vlan 30
Switch(config-vlan)# name VLAN30
Switch(config)# vlan 50
Switch(config-vlan)# name VLAN50
Switch(config)# interface range fa0/1-3
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 10
Switch(config)# interface range fa0/4-6
```

```

Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 30
Switch(config)# interface range fa0/7-9
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 50
Switch(config)# interface fa0/10
Switch(config-if)# switchport mode trunk
Switch(config-if)# switchport trunk allowed vlan 10,30,50

```

Verification: a ping between two PCs in the same VLAN on different switches (for example PC18 on Switch6 and PC27 on Switch7) succeeds, because the frames cross the trunk with an 802.1Q tag. A ping between PCs in different VLANs (for example PC18 and PC21) times out, because there is no Layer 3 device to route between the subnets.

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC24	PC35	ICMP		0.000	N	0	(edit)	
	Failed	PC18	PC21	ICMP		0.000	N	1	(edit)	

Figure 9 – Same-VLAN ping succeeds; cross-VLAN ping fails

3.7 NAT-PAT Edge Configuration (Question 6)

Router6 sits at the edge of the private network (Switch8 and PC36) and Router7 represents the outside network. Router6's inside interface (Gi0/0) faces the private LAN 172.25.1.0/27, and its outside interface (Gi0/1) faces Router7 over the 172.25.1.128/30 link. Router7 has an outside LAN 172.25.1.96/29 (Switch9) holding Server14, Server15 and PC37, and it does not run NAT.

Device	Interface / host	IP address	Role
Router6	Gi0/0	172.25.1.1/27	Inside (private) gateway
Router6	Gi0/1	172.25.1.129/30	Outside; PAT overload address
Router7	Gi0/0	172.25.1.130/30	Link to Router6
Router7	Gi0/1	172.25.1.97/29	Gateway of outside LAN
Server14 / Server15 / PC37	Fa0	172.25.1.98 / .99 / .100	Outside hosts
PC36	Fa0	172.25.1.2 – 172.25.1.30 (DHCP)	Private host

```

! Router6
Router6(config)# interface gi0/0
Router6(config-if)# ip address 172.25.1.1 255.255.255.224
Router6(config-if)# ip nat inside
Router6(config-if)# no shutdown
Router6(config)# interface gi0/1
Router6(config-if)# ip address 172.25.1.129 255.255.255.252
Router6(config-if)# ip nat outside
Router6(config-if)# no shutdown
Router6(config)# access-list 1 permit 172.25.1.0 0.0.0.31
Router6(config)# ip nat inside source list 1 interface gi0/1 overload
Router6(config)# ip route 0.0.0.0 0.0.0.0 172.25.1.130
! Router7
Router7(config)# interface gi0/0
Router7(config-if)# ip address 172.25.1.130 255.255.255.252
Router7(config-if)# no shutdown
Router7(config)# interface gi0/1
Router7(config-if)# ip address 172.25.1.97 255.255.255.248
Router7(config-if)# no shutdown

```

Access list 1 selects the private addresses to translate. The overload keyword makes this PAT: many inside hosts share the single outside address 172.25.1.129 and are told apart by source port number. Without overload, a separate outside address would be needed for each inside host. Router7 needs no extra route, because the translated source address 172.25.1.129 is directly connected to it. The 172.25.1.128/30 link and the outside LAN play the role of the public network in this lab, since the brief requires every address to come from the assigned block.

Verification: a ping or web request from PC36 to Server14 succeeds, and show ip nat translations on Router6 shows PC36's private address mapped to 172.25.1.129 with a port number. Show ip nat statistics shows the translation counters.

```
C:\>ping 172.25.1.98

Pinging 172.25.1.98 with 32 bytes of data:

Reply from 172.25.1.98: bytes=32 time<1ms TTL=126
Reply from 172.25.1.98: bytes=32 time=1ms TTL=126
Reply from 172.25.1.98: bytes=32 time<1ms TTL=126
Reply from 172.25.1.98: bytes=32 time<1ms TTL=126

Ping statistics for 172.25.1.98:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Figure 10 - Ping from PC36 to Server14

4. Discussion

VLSM was applied to the assigned /23 block so that each network received a block matching its need instead of a fixed size. A sizing detail that changes the result is the extra devices: Router0 and Router5 host the DNS and web servers, so their subnets are larger than the raw host counts suggest (Router0 needs a /25, not a /26). VLAN 10 needs exactly 30 hosts and uses a /27, which has exactly 30 usable addresses; this works because the VLAN has no gateway address to reserve.

The project text asks for six router-to-router link blocks. The supplied Packet Tracer topology connects six routers in a chain, which uses five links, so the addressing plan reserves a sixth /30 for a Router5-Router0 link. The design follows the supplied file and leaves that link uncabled. The consequence is that the chain has no redundant path: if any single core link fails, the routers on the far side lose contact with the rest. Cabling the reserved link and advertising it in OSPF would turn the chain into a ring and let OSPF reroute around a failure.

The DNS design uses two servers, one at each end of the chain, so clients on either side have a nearby server. Each DNS server was given records for all six websites, so every client can reach every site regardless of which server it was assigned by DHCP. Because OSPF provides reachability between all LANs, this split is a design choice, and a single DNS server would also have worked.

The VLAN segment is deliberately Layer 2 only. VLAN isolation here comes from the absence of any router: devices in the same VLAN reach each other across the trunk, while different VLANs are separate broadcast domains with no path between them. Adding inter-VLAN routing (router-on-a-stick) would defeat the isolation requirement.

NAT-PAT was verified with traffic generated from the private network and by inspecting the translation table. It confirms that inside hosts appear to the outside network with one shared address, which is how a real organisation conserves public IPv4 addresses. In this lab the outside side uses addresses from the assigned block, so the "public" address is a stand-in for a real public address.

Every function was tested rather than only configured: DHCP by checking the addresses received, DNS and web by opening sites by name, OSPF by checking neighbours, routes and pings between LANs, VLANs by same-VLAN and cross-VLAN pings, and NAT-PAT by the translation table. The screenshots in this report come from the submitted Packet Tracer file.

6. Conclusion

The assigned 172.25.0.0/23 block was divided with VLSM into eleven LAN and VLAN subnets and seven point-to-point links (five in use, one reserved and the NAT link), using 388 of the 512 addresses and leaving 124 free for growth. On this addressing plan, dedicated DHCP servers configure the hosts in every branch, two DNS servers and six web servers provide name-based access to six websites, and OSPF gives full reachability across the six-router chain. A two-switch VLAN segment keeps three groups isolated at Layer 2, and a NAT-PAT border between Router6 and Router7 shares a single address among the private hosts. Each requirement of the project was implemented in Packet Tracer and checked with the tests described above.

7. References

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